DA4022 MID-EVALUATION REPORT

Project 7: Uncertainty for Optimal Memory Allocation

PROJECT TEAM MEMBERS
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1. Background

Large Language Models (LLMs) are increasingly deployed in settings where computational resources are limited. A key bottleneck is how memory and compute are allocated during inference — particularly Key-Value (KV) cache size and generation strategy (greedy decoding vs. beam search). Currently, these execution parameters are statically fixed regardless of how confident the model is in its generated output.

Two recent ICLR 2026 papers address uncertainty estimation in LLMs from complementary angles:

- **Paper 1 — Query-Level Uncertainty (Internal Confidence):** Proposes estimating model uncertainty *before* generating any tokens, via a single forward pass. The method, called Internal Confidence (IC), probes `P(YES)` logits at every (layer, token) position and aggregates them using Attenuated Encoding centered at the last layer and last token. This produces a cheap, training-free uncertainty scalar that predicts whether the model can correctly answer a query.
- **Paper 2 — SemanticDPP:** Proposes estimating uncertainty *after* sampling a small set of responses, by measuring their semantic diversity using a Determinantal Point Process (DPP) over internal model embeddings. A learned projection maps embeddings to a semantic subspace, and the log-determinant of the resulting kernel matrix captures how diverse (uncertain) the responses are. An extension, SemanticDPP-C, uses greedy Maximum A Posteriori (MAP) clustering to identify the number of semantically distinct clusters.

Crucially, **neither paper connects uncertainty to downstream memory or compute allocation**.

2. Research Gap and Novel Contribution

2.1 What the Papers Leave Open

Aspect	Paper 1 (IC)	Paper 2 (SemanticDPP)
Uncertainty type	Pre-generation scalar	Post-generation diversity
Computation cost	~1 forward pass (zero decode)	m forward passes (m ≥ 3)
Output	Single IC score ∈ [0, 1]	log-det scalar + kernel matrix
Downstream use	Binary: trigger RAG / abstain	Binary: abstain threshold
Memory allocation	✗ Not addressed	✗ Not addressed
Adaptive compression	✗ Not addressed	✗ Not addressed

> 🎯 **The Core Research Gap**
> Both papers produce uncertainty signals but use them solely as binary on/off switches. Neither framework uses uncertainty as a *continuous control variable* to dynamically modulate how much memory or compute is allocated per query.

2.2 Our Novel Contribution

We propose **Uncertainty-Calibrated Adaptive Compression (UCA)**: a system that leverages IC as a zero-cost pre-filter and exploits the DPP eigenspectrum as a richer, per-dimension uncertainty signal to continuously control KV cache budget and generation strategy.

Three specific contributions beyond both papers:

1. **IC-Gated DPP Cascade:** IC pre-filters queries. Only uncertain queries (~40–50%) proceed to the more computationally demanding DPP computation, saving roughly half the overall DPP overhead.
2. **DPP Eigenspectrum as Memory Oracle:** Paper 2 computes L_S but discards all per-dimension structure, retaining only $\log\det(L_S)$. We expose the full eigenspectrum $\{\sigma_1, \ldots, \sigma_m\}$ of L_S and map each eigenvalue to an individual per-dimension compression ratio:

   > $r_i = \text{clip}(1 - \alpha \cdot \sigma_i / \sigma_{\max},\ r_{\min},\ r_{\max})$

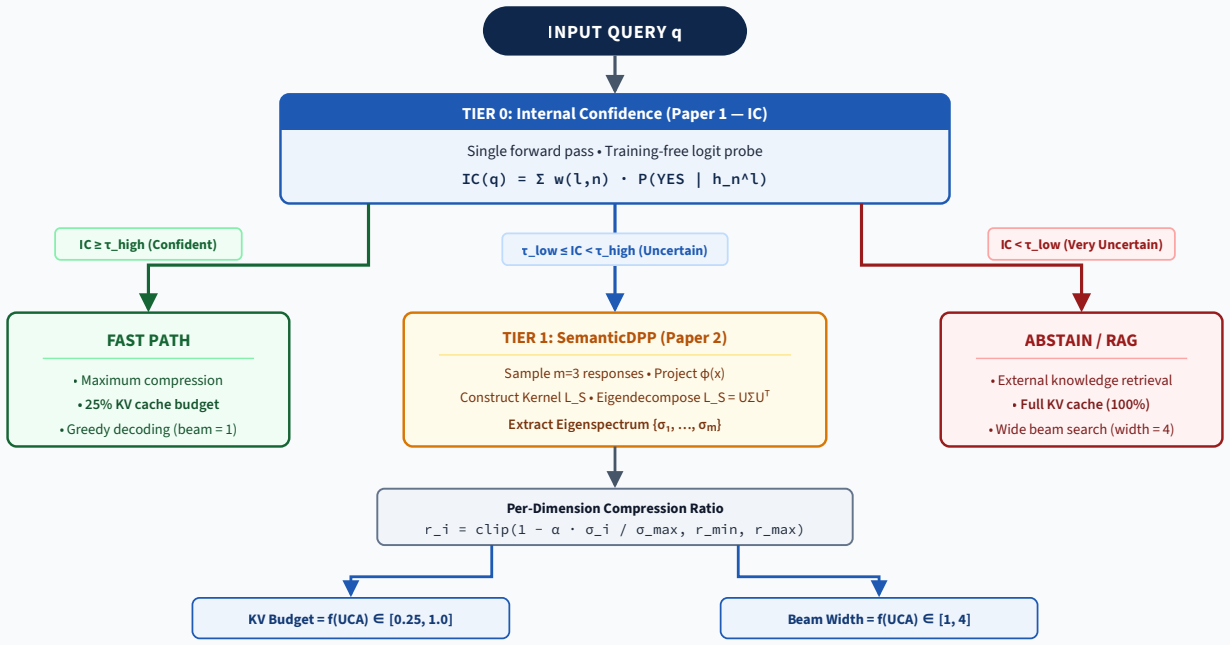
   A high eigenvalue σ_i indicates a contested semantic dimension requiring higher fidelity (less compression), whereas a low σ_i permits aggressive compression.

3. **Continuous KV Budget Control:** The composite UCA score drives KV cache budget and beam width continuously, replacing rigid heuristic selections:

> $\text{UCA} = 0.4 \cdot (1 - \text{IC}) + 0.6 \cdot \text{DPP_uncertainty}$ | $\text{KV_budget} = 0.25 + 0.75 \cdot \text{UCA}$

3. Proposed Architecture

FIGURE 1: COMPLETE UCA INFERENCE & DYNAMIC ALLOCATION PIPELINE



Key Design Decisions Grounded in Literature:

- $\tau_{high} = 0.65$, $\tau_{low} = 0.30$ — Grounded directly in IC's reported adaptive RAG threshold experiments (Paper 1, Section 5).
- $\alpha = 1.0$ in Attenuated Encoding — Kept fixed per Paper 1, Section 4.1 for training-free execution.
- $m = 3$ DPP responses — Paper 2 uses $m = 5$ for peak standalone AUROC; we use $m = 3$ since the IC pre-filter removes most clear queries.
- DPP-C cluster count `k = max(1, round(trace(K)))` — Follows Paper 2, Algorithm 1; utilized directly as an adaptive beam width floor.

4. Implementation Progress

4.1 Completed DONE

- **Environment Setup:** Python 3.11, PyTorch 2.3, HuggingFace Transformers, and the official `query-level-uncertainty` package installed and verified.
- **IC Gate (`src/core/internal_confidence.py`):** Implemented the `ICGate` class wrapping the official `tigerchen52/query_level_uncertainty` package. Self-evaluation prompt template and Attenuated Encoding with $\alpha = 1.0$ operational. Tested on TriviaQA with Llama-3.2-3B; scores route reliably into fast, uncertain, and abstain buckets.
- **SemanticDPP Module (`src/core/semantic_dpp.py`):** Implemented `SemanticSubspaceProjector` (orthogonal-initialized NN, $d \to 256$) and kernel matrix computation:

> $K(x_i, x_j) = \exp(-\alpha \, ||\phi(x_i) - \phi(x_j)||^2)$

  The `uncertainty_score()` returns `log_det`, normalized uncertainty, eigenspectrum, and DPP-C cluster count. Eigenvalue extraction via `torch.linalg.eigvalsh` is numerically verified.
- **UCA Controller (`src/core/uca_controller.py`):** Wires IC and DPP together. Method `compute_compression_config()` produces execution route, KV budget, beam width, and per-dimension compression ratios for any incoming query. Unit-tested on sample inputs; routing and continuous control logic verified.

4.2 In Progress ACTIVE

- **Projector Training:** Training `SemanticSubspaceProjector` by maximizing DPP marginal likelihood:

  > $\Delta = \log\det(L_Y) - \log\det(L + I)$

  on pseudolabeled TriviaQA clusters. Optimization loop is implemented; tuning hyperparameters for stable convergence.
- **Baseline IC Reproduction:** Running IC benchmark on TriviaQA validation split (Llama-3.2-3B) to empirically reproduce published AUROC numbers. Evaluation is currently underway.

4.3 Not Yet Started PLANNED

- **Full UCA End-to-End Evaluation (`run_uca.py`):** Integrating routing with KV cache compression execution.
- **SemanticDPP Standalone AUROC Measurement:** Running full sampling-based evaluation without the pre-filter.
- **Ablation Experiments:** Isolating individual contributions of eigenspectrum and gating mechanisms.

5. Evaluation Plan

We will evaluate UCA against two standalone baselines drawn directly from the two reference papers.

Evaluated Baselines

- **IC Only** (Paper 1 standalone): IC score alone for routing; no DPP stage.
- **SemanticDPP Only** (Paper 2 standalone): DPP for every query; log-det scalar threshold for abstain.

Benchmark Datasets & Models

- **Datasets:** TriviaQA (factual retrieval), GSM8K (multi-step reasoning) — matching Paper 1 protocol.
- **Base Model:** Llama-3.2-3B.

Primary Evaluation Metrics

- **AUROC:** Uncertainty score vs. correctness ground truth (primary discriminative metric from both papers).
- **PRR (Prediction Rejection Ratio):** Measures how rapidly error rate drops as high-uncertainty queries are rejected.
- **Average KV Tokens Used:** Measures physical GPU memory savings achieved through adaptive budgeting.
- **End-to-End Latency per Query:** Profiling overhead — IC adds ~1 forward pass; DPP adds ~3 generation passes.

References

[1] Chen, L., de Melo, G., Suchanek, F. M., & Varoquaux, G. (2026). *Query-Level Uncertainty in Large Language Models*. International Conference on Learning Representations (ICLR 2026). https://openreview.net/pdf?id=11QZITAMUO

[2] Hamer, J., Snoek, J., et al. (2026). *SemanticDPP: Efficient Uncertainty Quantification in LLMs*. International Conference on Learning Representations (ICLR 2026). https://openreview.net/pdf?id=C8yjlrx4Wo